Three robots.
Local AI control.
Measured comparison.

A 3D integration demonstration with a preliminary held-out study. Prepared October 25, 2026.



Task and purpose

Three robots start simultaneously and travel to individually marked goals, avoiding static obstacles and one another. The study compares a classical navigation reference, behavior cloning (BC) and parameter-shared independent PPO. Shared-space transport is the motivating application; this is simulation research, not a deployed warehouse system.

Main result: PPO completed 20/40 held-out 2D episodes, versus 12/40 for classical and 10/40 for BC. All three controllers completed one easy 3D integration race without contact. The 80% success target was not met.

01 / System and AI methods

Common controller contract

Each robot receives 44 local values: goal distance/bearing, own speed/turn rate, 24 planar ranges and up to three nearby peers with relative pose/motion and presence masks. Outputs are forward speed [0, 1] m/s and angular speed [-2, 2] rad/s. The controller interval is 0.1 s. Each method shares an explicit 0.27 m arrival-stop rule, with goal tolerance 0.3 m and five stopped dwell steps. Finished robots remain in the world.

Method	Training and inference
Classical	Goal steering, range avoidance and local safety/yielding heuristic. No learning.
BC	Bounded-action MLP with 128/128/64 hidden units. 4,057 fit samples; 1,041 internal holdout samples. Whole-episode split. Train-only normalization. 50 epochs; validation selected epoch 40.
Independent PPO	Shared 64/64 actor and critic; local observations. Bounded Beta policy trained from scratch with PPO/GAE. 32,768 world steps; 91,215 policy-agent transitions. Validation selected update 96. Deterministic mean-action evaluation.

3D integration

PyBullet DIRECT advances 1 kg cylindrical chassis with radius 0.2 m and height 0.24 m. An ideal body-frame velocity servo runs at 0.01 s physics steps with gravity disabled. Wheels and heading markers are visual only: tire friction, wheel torque, suspension and actuator dynamics are not modeled. Contacts come from Bullet; measured poses feed the existing observation adapter. Planar sensor intersections use matching cylinder/box footprints, rather than Bullet ray casts. There is no classical obstacle-avoidance fallback in learned inference.

Original implementation contributions

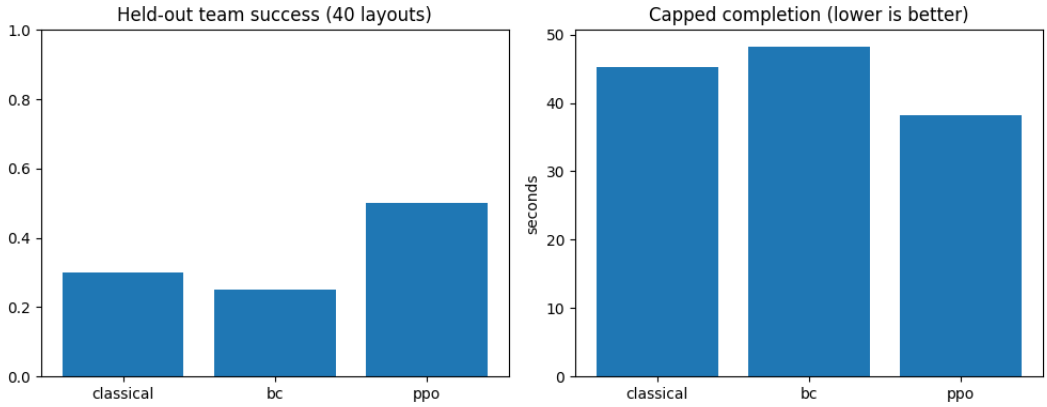
The project combines deterministic multi-robot scenarios, a common local observation/action interface, custom BC/PPO pipelines, a 3D physics adapter, contact-aware race logging, raw camera capture and paired evaluation. Three colored chassis, raised obstacles, start/goal markers, finish timing and explicit ties make the race traceable. External libraries provide numerical computing, learning, physics, image/video encoding and PDF layout; no external robot artwork was used.

Retained pilot compute: BC fitting 24.16 s plus expert collection 126.31 s; PPO training 392.12 s on CPU. These are unequal pilot budgets and do not constitute a complete historical tuning-cost audit. Training seed 0 only for each learned method.

02 / Evaluation and results

The frozen preliminary test uses 40 paired, previously held-out 2D layouts: ten each from open, slalom, bottleneck and crossing. The first ten of each family were selected from the existing 100-layout manifest before running methods. Checkpoints were selected on validation only. No additional training or test-driven retuning was performed.

Controller	Success	Collisions	Timeouts	Capped s	Success-only s
Classical	12/40	0	28	45.195	10.650
BC	10/40	30	0	48.260	13.040
PPO	20/40	20	0	38.213	16.425



Interpretation and uncertainty

PPO is the practical AI-demo recommendation because it has the highest team success and lowest failure-capped completion time. Failure contributes the full 60 s horizon, preventing early collisions from appearing fast. Successful makespan is conditioned on completion. All methods achieved 0/10 successes on both slalom and bottleneck. PPO completed all open and crossing episodes; BC completed only open; classical completed open plus two crossing episodes.

Paired scenario bootstrap: 5,000 resamples, seed 415. The 95% success-difference intervals are classical minus BC [0.000, 0.125], classical minus PPO [-0.325, -0.075], and BC minus PPO [-0.400, -0.125]. These intervals cover scenario sampling only. One training seed gives no training-variation estimate. CPU contention during dependency compilation limits inference-latency comparisons.

3D transfer: a separate integration result

On predesignated scene seed 41000, all three robots finished without contact for each controller: classical 10.2 s, BC 10.4 s and PPO 15.9 s team makespan. PPO robot finish times were [15.7, 15.8, 15.9] s. The scene has two raised obstacles between separated lanes; it is an easy integration scene. Matching 2D reference makespans were 10.2/10.4/15.9 s. These results are not pooled with the held-out comparison and do not establish obstacle-avoidance reliability.

03 / Evidence and reproduction

Submission artifacts

The package contains a captioned 120-second MP4, this report, source/configs, selected BC/PPO weights, training summaries and curves, scenario manifests, raw episode results, raw 3D videos and step logs. The film includes an uninterrupted 16-second PPO race at 1x, a synchronized three-method view with labeled terminal holds, comparison metrics and a clearly identified 2D failure still. No personal account identifier is embedded in the media.

Run from the extracted code root

Use Windows x64 with Python 3.13. Create a virtual environment and install requirements-release.txt (CPU Torch index included). Then run:

```
python -m pytest -q
python -m roborace.g4 demo
python -m roborace.g4 compare
```

The demo loads included checkpoints and writes three 3D MP4s plus logs. The compare command reproduces the fixed 40-layout study; do not tune policies against these consumed test layouts. Training commands and configurations are included. The release verification record states the actual clean-environment checks, without implying a full training rerun.

Evidence index and attribution

Evidence	Package path
Frozen comparison / raw records	artifacts/g4/frozen_manifest.json; episodes.jsonl; summary.json
3D model / logs / provenance	artifacts/g4b/backend.json; runtime.json; scene.json; */steps.jsonl; */episode.json
Training / selected checkpoints	artifacts/g2/bc_seed0/; artifacts/g3/ppo_seed0/
Source / dependencies / scenarios	roborace/; scripts/; configs/; requirements-*.txt; scenarios/
Integrity / third-party notices	SHA256SUMS.json; THIRD_PARTY_NOTICES.md; package license files

Libraries: NumPy, PyTorch, PyBullet, Pillow, Matplotlib, imageio/imageio-ffmpeg and ReportLab. Package notices preserve dependency license metadata; no course sample assignment code is bundled. Course-provided assignment PDFs and private planning documents are excluded from the public-ready code archive. The source was developed with AI assistance.

Limits and release status

Full multi-seed G4 and wider G4b transfer validation remain incomplete. Minimum-clearance/path-efficiency analysis and realistic wheel dynamics are not implemented. Local G5 media and packaging do not complete course submission. Account-specific naming, public GitHub access, YouTube upload/settings, current portal confirmation and OmniCampus submission must be verified separately. No external publication or submission is claimed.